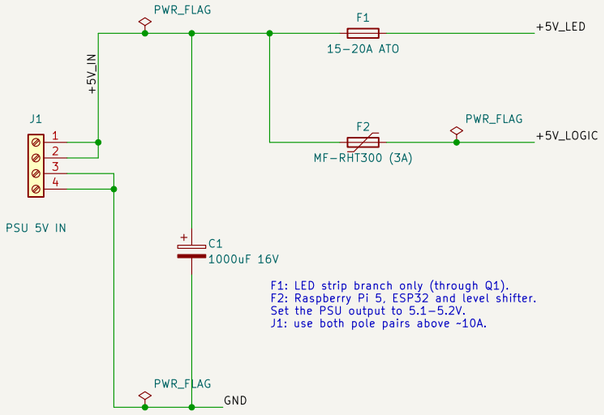
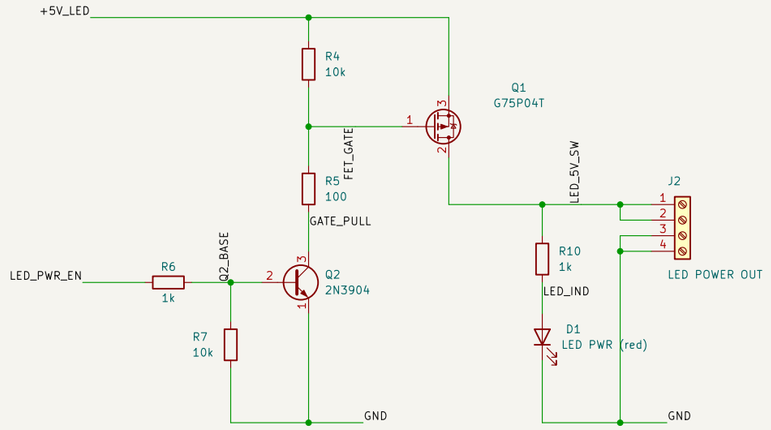
POWER INPUT 5V (10–20A PSU)



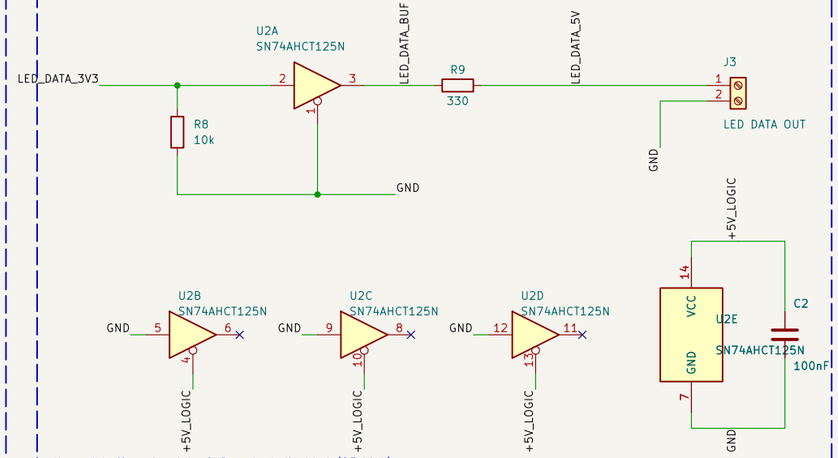
F1: LED strip branch only (through Q1).
F2: Raspberry Pi 5, ESP32 and level shifter.
Set the PSU output to 5.1–5.2V.
J1: use both pole pairs above ~10A.

LED POWER SWITCH (high side, active-high GPIO16)



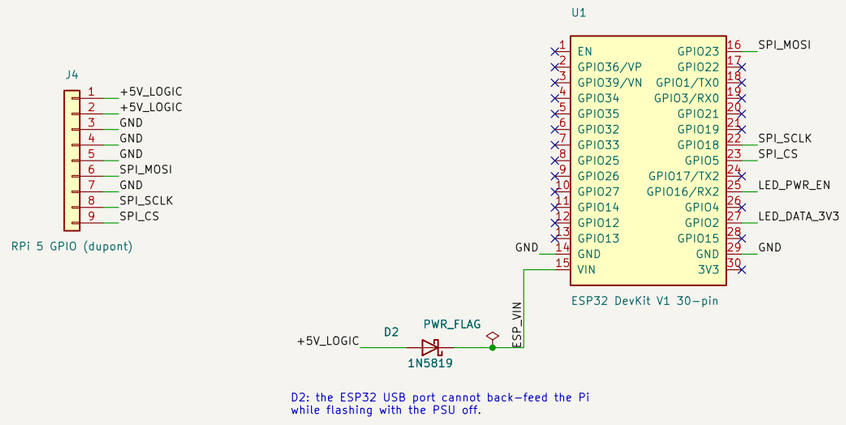
GPIO16 HIGH -> Q2 on -> Q1 gate low -> LED power ON.
R4/R7 keep Q1 OFF while the ESP32 boots (GPIO16 floats).
Firmware: -DLED_POWER_PIN=16, no -DLED_POWER_INVERT.
Q1: logic-level P-MOSFET, Vds >= 20V, Rds(on) <= 10 mOhm at Vgs = -4.5V. Add a TO-220 heatsink above ~10A.

LED DATA LEVEL SHIFTER 3.3V -> 5V



Unused buffers: input to GND, output disabled (OE high).
C2 next to U2 pin 14. R9 close to J3.
R8 keeps the data line low while the ESP32 boots.

ESP32 DEVKIT + RASPBERRY PI 5 (dupont cables)



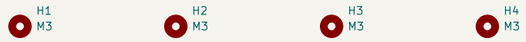
D2: the ESP32 USB port cannot back-feed the Pi while flashing with the PSU off.

J4 -> Raspberry Pi 5 header
1 +5V -> pin 2 6 MOSI -> pin 19 (GPIO10)
2 +5V -> pin 4 7 GND -> pin 20
3 GND -> pin 6 8 SCLK -> pin 23 (GPIO11)
4 GND -> pin 9 9 CE0 -> pin 24 (GPIO8)
5 GND -> pin 14
Short 22AWG dupont wires; keep both 5V and all GND wires.

NOTES

1. All components are through-hole. U1 plugs into two 1x15 female headers, U2 into a DIP-14 socket.
2. Measure the ESP32 header row spacing before the PCB layout (footprint assumes 25.4 mm) and check the board silkscreen matches the U1 pin labels (DOIT DevKit V1 30-pin layout).
3. High-current path: J1 -> F1 -> Q1 -> J2. Use wide copper pours, 2 oz copper, and both poles of J1/J2 in parallel above ~10A. Use 14AWG (12AWG at 20A) wire. Each 5.08 mm terminal pole is rated ~13.5A. Long strips: inject power at both ends, taken from J2.
4. One common ground: PSU, LED strip, ESP32 and Raspberry Pi. Run the J3 GND wire alongside DATA.
5. LED data: GPIO2 -> U2A (SN74AHCT125N, must be AHCT) -> R9 330 Ohm -> strip DIN.
6. Firmware: pio run -e WS281x_RGB_LED_POWER (LED_POWER_PIN=16, active-high, no LED_POWER_INVERT).
7. Q1: Goford G75P04T (-40V, Rds(on) max 9 mOhm @ Vgs -4.5V, pins G-D-S). Any TO-220 P-MOSFET specified for Vgs = -4.5V also works. IRF4905/IRF9540 are NOT suitable (not logic-level).
8. Raspberry Pi 5 is powered through GPIO 5V pins: its USB ports are limited to 600 mA unless usb_max_current_enable=1 is set in config.txt (only if the PSU can supply it).
9. HyperHDR: keep Smoothing enabled so static images are resent (inactivity timeout).

MOUNTING



All components through-hole. 5.08 mm screw terminals for power and LED outputs.
ESP32-WROOM-32 SPI slave, P-MOSFET LED power switch on GPIO16, AHCT125 level shifter

HyperSPI (ESP32) + Raspberry Pi 5 + HyperHDR

Sheet: /
File: hyperspi-led-controller.kicad_sch

Title: HyperSPI LED Controller

Size: A3	Date: 2026-10-21	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 1/1